

Assignment 1: Student Login and Dashboard System Using Session

Problem Statement

Design and develop a simple **Student Login and Dashboard System** using Java Servlets, HTML, JSP, and Session Management.

The application should allow a student to log in using valid credentials. After successful login, the student should be able to access the dashboard. If the student is not logged in and tries to access the dashboard directly, the system should redirect the student back to the login page.

This assignment focuses on understanding how sessions are used to maintain user login state across multiple requests.

Functional Requirements

The application should contain the following pages and components:

Component	Purpose
Login Page	Allows student to enter username and password
Login Processing Servlet	Validates login details
Dashboard Page	Displays welcome message after successful login
Logout Servlet	Destroys the session and logs the user out

Login Rules

Use fixed credentials for login validation.

Username	Password
student	student123

If the credentials are correct, the system should create a session and store the logged-in student's username.

If the credentials are incorrect, the student should be redirected back to the login page.

Dashboard Rules

The dashboard should be accessible only after successful login.

If a user directly opens the dashboard URL without logging in, the application should not allow access.

The dashboard should display:

Details to Display

Welcome message

Logged-in student username

Logout option

Logout Rules

When the student clicks logout:

Expected Behaviour

The session should be destroyed

The user should be redirected to the login page

The dashboard should not be accessible after logout

Expected Application Flow

```
login.html
  ↓
LoginServlet
  ↓
If valid → create session → dashboard
If invalid → redirect to login.html

dashboard
  ↓
If session exists → show dashboard
If session does not exist → redirect to login.html

logout
  ↓
Destroy session
  ↓
Redirect to login.html
```

Assignment 2: Online Shopping Cart Using Session

Problem Statement

Design and develop a simple **Online Shopping Cart System** using Java Servlets, HTML, JSP, and Session Management.

The application should allow a user to select products and add them to a shopping cart. The cart should remember selected products across multiple requests using session. The user should be able to view the cart and remove all items from it.

This assignment focuses on how session can be used to store user-specific temporary data.

Functional Requirements

The application should contain the following components:

Component	Purpose
Product List Page	Displays available products
Add to Cart Servlet	Adds selected product to session cart
View Cart Page	Displays all products added to cart
Clear Cart Servlet	Removes all products from the cart

Product List

The product list should contain at least five products.

Example:

Product ID	Product Name	Price
P101	Keyboard	800
P102	Mouse	500
P103	Monitor	7000
P104	USB Cable	250
P105	Laptop Stand	1200

Cart Rules

When the user adds a product to the cart:

Rule

The selected product should be stored in session

Multiple products should be allowed

Cart data should remain available while the session is active

Cart should be user-specific

Cart should be cleared when the user chooses clear cart

View Cart Requirements

The cart page should display:

Details to Display

Product name

Product price

Total number of products

Total cart amount

If the cart is empty, the system should display a proper message.

Clear Cart Requirements

When the user clears the cart:

Expected Behaviour

All cart items should be removed from the session

User should be redirected to the product list or cart page

Cart should show empty message after clearing

Expected Application Flow

```
product-list.html / product-list.jsp
↓
User selects product
↓
AddToCartServlet
↓
Product stored in session
↓
Redirect to product list or cart page

view-cart
↓
Read cart from session
↓
Display selected products

clear-cart
↓
Remove cart from session
↓
Show empty cart
```

Assignment 3: Remember Username Using Cookies

Problem Statement

Design and develop a simple **Remember Username System** using Java Servlets, HTML, JSP, and Cookies.

The application should allow a user to enter their username and choose whether the browser should remember it. If the user selects the remember option, the username should be stored in a cookie. When the user visits the login page again, the saved username should be displayed or pre-filled.

This assignment focuses on using cookies to store small non-sensitive client-side data.

Functional Requirements

The application should contain the following components:

Component	Purpose
Login Page	Allows user to enter username
Remember Username Option	Allows user to choose whether username should be saved
Cookie Creation Servlet	Stores username in cookie
Login Page Revisit	Displays saved username if cookie exists
Delete Cookie Option	Allows user to remove saved username

Form Requirements

The login form should contain:

Field

Username

Password

Remember Username checkbox

Only the username should be stored in the cookie.

The password must not be stored in a cookie.

Cookie Rules

If the user selects “Remember Username”:

Rule

Username should be stored in a cookie

Cookie should remain available even after browser restart

Cookie should have a fixed expiry time

If the user does not select “Remember Username”:

Rule

Username should not be stored permanently

Any existing saved username cookie should be removed

Delete Cookie Requirement

The application should provide an option to remove the saved username.

After deletion:

Expected Behaviour

Saved username should not appear again

Cookie should be removed from the browser

User should be redirected to the login page

Expected Application Flow

```
login page
↓
User enters username
↓
User selects Remember Username
↓
Servlet creates cookie
↓
User revisits login page
↓
Saved username is displayed
```

delete saved username



Cookie is removed



Login page opens without saved username

Assignment 4: User Preference System Using Cookies

Problem Statement

Design and develop a simple **User Preference System** using Java Servlets, HTML, JSP, and Cookies.

The application should allow a user to select personal preferences such as theme and language. These preferences should be saved in cookies so that when the user visits the application again, the same preferences are automatically applied.

This assignment focuses on using cookies for storing small user preferences on the client side.

Functional Requirements

The application should contain the following components:

Component	Purpose
Preference Form Page	Allows user to select preferences
Save Preference Servlet	Stores selected preferences in cookies
Home Page	Displays content based on saved preferences
Reset Preference Servlet	Deletes preference cookies

Preference Options

The user should be able to select:

Preference	Options
Theme	Light, Dark
Language	English, Hindi, Marathi
Font Size	Small, Medium, Large

Cookie Rules

When the user saves preferences:

Rule

Theme should be stored in a cookie

Language should be stored in a cookie

Font size should be stored in a cookie

Preferences should remain available for future visits

Cookies should have a fixed expiry time

Home Page Requirements

The home page should read the saved cookies and display the selected preferences.

Example output:

Display

Selected Theme: Dark

Selected Language: English

Selected Font Size: Medium

If no cookies are found, default preferences should be shown.

Default preferences:

Preference	Default Value
Theme	Light
Language	English
Font Size	Medium

Reset Preference Requirements

The application should provide an option to reset preferences.

After reset:

Expected Behaviour

Preference cookies should be deleted

Default preferences should be applied

User should be redirected to the home page

Expected Application Flow

preference-form.html

↓

User selects theme, language, font size

↓

SavePreferenceServlet

↓

Preferences stored in cookies

↓

Redirect to home page

home page

↓

Reads cookies

↓

Applies saved preferences

reset preferences

↓

Deletes preference cookies

↓

Applies default preferences